

# HW04 - AI Automation Testing Report

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<https://github.com/linhnph05/hw04---Automation-testing>

## 1. Scope and feature selection

I tested these three web features:

- Pool A - FR-03: Forgot password and reset password.
- Pool B - FR-09: Discount coupons.
- Pool C - FR-12: Admin access control.

## 2. Method

I used AI step by step. First, I asked a Codex agent to read the requirements and source code. Next, I asked it to design 12 test cases for each feature. I reviewed the cases before asking for a Playwright draft for each feature. I changed the drafts from CommonJS to ES modules, fixed incorrect assumptions, and ran the tests in Chromium. I then asked the AI to review its drafts and made a final check myself.

## 3. Automated cases and assertion patterns

Each JSON file contains 12 test cases. There are 36 cases in total. Running them in three browsers produces 108 test executions.

All 36 cases are written in a simple text format in TEST\_CASES.md (TEST\_CASES.md). This file shows the setup, input, actions, and expected result. Playwright reads the test data from the JSON files.

The suite uses more than three assertion patterns, including:

1. Check visibility and item count with `toBeVisible()` and `toHaveCount()`.
2. Check exact values and status codes with `toBe()`.
3. Check text with `toMatch()` and `toContainText()`.
4. Check the current page with `toHaveURL()`.
5. Check object values with `toMatchObject()`.
6. Check rejected actions with `not.toEqual()` and confirm that blocked data was not saved.

### How to run the tests

Start the backend, customer website, and admin website on ports 3000, 5173, and 5174. Then install the project packages and browsers:

```
npm install
npx playwright install chromium firefox webkit
```

Run one feature in Chromium:

```
npx playwright test tests/fr03.spec.js --project=chromium
```

Run all tests in one browser:

```
npx playwright test --project=chromium
```

Run all 36 cases in Chromium, Firefox, and WebKit:

```
npm run test:matrix
```

Some tests fail because they find real SUT defects. Playwright saves the new HTML report in `playwright-report/`.

## 4. Execution results

| Feature | Chromium            | Firefox | WebKit  | Total passed | Total failed |
|---------|---------------------|---------|---------|--------------|--------------|
| FR-03   | 7 passed / 5 failed | 7 / 5   | 7 / 5   | 21           | 15           |
| FR-09   | 7 passed / 5 failed | 7 / 5   | 7 / 5   | 21           | 15           |
| FR-12   | 5 passed / 7 failed | 5 / 7   | 5 / 7   | 15           | 21           |
| Total   | 19 / 17             | 19 / 17 | 19 / 17 | 57           | 51           |

The results are the same in all three browsers. This means the failures are likely SUT defects, not browser problems. I opened all nine HTML reports and checked that each report shows Run by: 23127081 and an ISO timestamp.

## 5. Human review of AI-generated drafts

| AI draft problem                                          | My correction                                             | Why the AI missed it                                         |
|-----------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------|
| Used CommonJS require in an ES module project             | Changed the drafts to import and JSON files               | The AI used an example without checking the project settings |
| FR-03 TC-01 checked OTP length and step controls together | TC-01 now checks the OTP only; TC-02 checks the controls  | The AI combined two requirements into one case               |
| Used exact alert text to check password reset             | Accepted the alert and checked the URL and login result   | Alert text can change and is not the main result             |
| Guest coupon test matched the Login link in the menu      | Checked the redirect to the Login page instead            | The AI did not limit the locator to the coupon result area   |
| Used a deleted coupon as an inactive coupon               | Replaced it with an empty-code case                       | The API has no inactive-coupon setup                         |
| Sent user_id in normal coupon requests                    | Used the user from the JWT token                          | The AI copied unsafe client behavior                         |
| Checked user isolation and percentage math in one case    | TC-11 checks user isolation; TC-01 checks percentage math | One case had two different reasons to fail                   |
| FR-12 tests checked only the HTTP status                  | Checked that blocked data was not saved                   | The first draft did not check saved data                     |
| Used only Date.now() for fixture names                    | Added a counter to stop duplicate names                   | Two fixtures can be made in one millisecond                  |

## 6. Defects found

I created eight GitHub issues. Each issue describes one main defect, so I did not create the same issue again for every browser:

- FR-03: four-digit OTP; missing step/navigation/confirmation controls; valid password rejected.
- FR-09: incorrect percent math; wrong equality boundary; missing authentication and trusted body user ID.
- FR-12: missing admin-role checks; unauthenticated product write APIs.

Issue links, screenshots, and test mappings are in BUG\_REPORT.md (BUG\_REPORT.md).

## 7. Report evidence

- HTML reports: reports/html/<feature>-<browser>/index.html.
- Visual summaries: images/\*-chromium-report.png.
- Failed tests include a Playwright trace and screenshot in the HTML report.

## 8. Conclusion

The final test suite uses separate data files and several assertion styles. It runs in Chromium, Firefox, and WebKit and produces nine HTML reports. The tests found important calculation and access-control defects. AI helped me read the code and create the first drafts, but I still had to fix the module format, selectors, test data, assertions, and cleanup.